NATHANAEL TEPAKBONG

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EDUCATION

Doctor of Philosophy - Department of Data Science Sep 2022–Aug 2026
City University of Hong Kong, Hong Kong SAR
- Hong Kong PhD Fellowship Scheme recipient
- Advisors: Prof. Xiang Zhou (CityU) & Prof. Ding-Xuan Zhou (University of Sydney)
- Thesis: Statistical Learning Theory for Provably Accurate and Efficient Scientific Machine Learning Algorithms

Master's Degree - Mathematical Research and Innovation 2020–2021
Université Paul Sabatier, Toulouse (France)
- Graduate level Probability Theory and Mathematical Statistics
- Relevant Coursework: Stochastic Calculus, Asymptotic Statistics, Statistical Learning

Master of Science - Aerospace Engineering ("Diplôme d'Ingénieur") 2017–2021
ISAE-Supaéro, Toulouse (France)
- Leading "Grande École" in Aerospace. Specialization in Applied Mathematics and Data Science
- Relevant Coursework: Advanced Statistics, Multi-Disciplinary Optimization, Algorithms in Machine Learning

Preparatory Classes - Mathematics, Physics and Computer Science 2015–2017
Lycée Buffon, Paris (France)

EXPERIENCE

City University of Hong Kong (Department of Mathematics), Hong Kong Sep 2026–present
Postdoctoral Researcher

· Advisors: Prof. Xiang Zhou and Prof. Bowen Li

City University of Hong Kong (School of Data Science), Hong Kong Jun 2021–Sep 2022
Research Assistant

· Advisor: Prof. Ding-Xuan Zhou

INRIA (Team SERENA), Paris Mar–Jul 2020
Research Intern

· Advisor: Géraldine Pichot

Robert Bosch Research and Technology Center, Singapore Mar–Sep 2019
R&D Intern

· Advisor: Stanley Eey

PUBLICATIONS AND MANUSCRIPTS

A Unifying Perspective on Sampling through Kernelized Stationary Equations (2026)
N. Tepakbong, Y. Cao, X. Zhou
Manuscript in preparation.

$H^2(\Omega)$ **A Priori Error Bounds for Boundary-Adapted PINNs Solving Elliptic Dirichlet Problems (2026)**
N. Tepakbong, J. Fan, X. Zhou, D.-X. Zhou
arXiv preprint, arXiv:2607.19167.

Taming the Loss Landscape of PINNs with Noisy Feynman–Kac Supervision: Operator Preconditioning and Non-Asymptotic Error Bounds (2026)
N. Tepakbong, H. Hu, C. Liu, X. Zhou
Proceedings of the 43rd International Conference on Machine Learning (ICML 2026), vol. 306. PMLR.

Super-Fast Rates of Convergence for Neural Networks Classifiers under the Hard Margin Condition (2026)
N. Tepakbong, X. Zhou, D.-X. Zhou
Transactions on Machine Learning Research (TMLR). OpenReview.

Initialization of the Circulant Embedding Method to Speed Up the Generation of Gaussian Random Fields (2022)
G. Pichot, S. Legrand, M. Kern, N. Tepakbong
The SMAI Journal of Computational Mathematics, 8, 327–347. doi:10.5802/smai-jcm.89.

THESES

Statistical Learning Theory for Provably Accurate and Efficient Scientific Machine Learning Algorithms (2026)
N. Tepakbong
PhD thesis, City University of Hong Kong. PDF.

Some Theory of Functional Data Classification with Shallow Feedforward Neural Networks (2021)
N. Tepakbong
Master's thesis, University of Toulouse and ISAE-Supaéro.

RESEARCH CONFERENCES

Frontiers in the Intersection of Stochastic Dynamical Systems and Machine Learning Sep 2026
Tianyuan Mathematics Research Center, Kunming, China

· Contributed talk: Sampling from invariant measures with gradient flows of the weak Fokker-Planck equation.

43rd International Conference on Machine Learning (ICML 2026) Jul 2026
COEX Convention and Exhibition Center, Seoul, South Korea

· Poster: Taming the Loss Landscape of PINNs with Noisy Feynman–Kac Supervision: Operator Preconditioning and Non-Asymptotic Error Bounds.

PhD Forum in Applied Mathematics Sep 2025
Fudan University, Shanghai, China

· Contributed talk: Super-Fast Rates of Convergence for Neural Networks Classifiers under the Hard Margin Condition.

The 14th AIMS Conference Dec 2024
New York University Abu Dhabi, Abu Dhabi, UAE

· Contributed talk: Solving for the Mean Escape Time with Physics-Informed Neural Networks: Theory and Applications.

International Conference on Applied Mathematics May 2024
City University of Hong Kong, Hong Kong SAR

· Contributed talk: Fast Asymptotic Rates of Convergence for Neural Networks under the Hard Margin Condition.

10th International Congress on Industrial and Applied Mathematics (ICIAM 2023) Aug 2023
Waseda University, Tokyo, Japan

· Poster: Exponential Convergence Rates for Binary Classification with Deep Neural Networks.

TECHNICAL SKILLS & LANGUAGES

Python (PyTorch, scikit-learn, pandas, matplotlib)	French - Native
C/C++, MATLAB, R	English - Fluent
CUDA, MPI, OpenMP	Mandarin - Conversational
Git, Bash, Docker, LaTeX	Cantonese - Conversational

TEACHING

SDSC2002 - Convex Optimization	Semester B 2025–26
SDSC5003 - Storing and Retrieving Data	Semester A 2025–26
SDSC2002 - Convex Optimization	Semester B 2024–25
SDSC3006 - Foundations of Machine Learning	Semester A 2023–24
SDSC2002 - Convex Optimization	Semester B 2023–24